

Predicting Elliptic Curve Ranks Using Frobenius Traces and Logistic Regression

Author: Nguyen Gia Khanh

Mentor: William DeGroot

Program: Inspirit AI

1. Abstract

This project investigates whether machine learning can distinguish elliptic curves of different ranks using their trace data. We ask the question: can logistic regression accurately classify rank 0 and rank 1 elliptic curves, and how does the model's accuracy depend on the conductor size of the training data? To study this, data from John Cremona's Elliptic Curve Database was used, containing over 10,000 curves labeled by rank. Logistic regression was trained on subsets of 2,000 curves (1,000 rank 0 and 1,000 rank 1) grouped by conductor size. The model was evaluated on accuracy for "low conductor," "high conductor," and combined datasets. We graphed many explicit examples of elliptic curve murmurations with conductors ranging from 0 to 10,000 and found that rank-separating signals weaken as conductor increases, which aligns with the model's declining accuracy across higher-conductor windows.

2. Introduction

Elliptic curves are central objects in modern number theory and cryptography, and their rank measures the number of rational points of infinite order. Understanding rank patterns computationally provides intuition for conjectures such as the Birch and Swinnerton-Dyer (BSD) conjecture.

For every prime p , an elliptic curve has a numerical invariant known as the Frobenius trace, denoted:

$$a_p = p + 1 - \#E(F_p)$$

where $\#E(F_p)$ represents the number of points the curve has over the finite field F_p . These traces encode rich arithmetic information about how the curve behaves modulo each prime.

The goal of this project is to use the Frobenius trace values of an elliptic curve to train a machine learning algorithm that can predict its rank. In this study, we apply a supervised learning model called logistic regression to datasets of elliptic curves that are labeled by rank. The main objective is to test whether patterns in the sequence of a_p values can be used to distinguish between curves of rank 0 and rank 1.

To better understand the behavior of Frobenius traces, mathematicians often use a visualization called an elliptic curve murmuration. In this plot, the x-axis represents a collection of primes, and the y-axis shows the average a_p value for a group of elliptic curves that share the same rank. These plots form smooth, oscillating wave patterns that differ between ranks, reflecting the underlying arithmetic structure of elliptic curves.

In this project, we focus primarily on distinguishing between rank 0 and rank 1 curves, with limited exploration of rank 2 curves. By combining ideas from mathematics and data science, we aim to show how machine learning can reveal meaningful patterns in elliptic curve data and provide new insight into the study of rank behavior.

3. Background

The idea for this project is motivated by a recent discovery called elliptic curve murmurations, first introduced in a 2022 research paper titled “*Elliptic Curve Murmurations*” (<https://arxiv.org/pdf/2204.10140>). In that study, the authors, including Andrew R. Booker and Andrew M. Sutherland observed that when the

Frobenius traces of many elliptic curves were averaged over primes, distinct oscillating wave patterns appeared. Each rank produced a different wave shape, which the authors called a murmuration. These waves revealed a surprising regularity in how elliptic curves behave statistically when grouped by rank.

The authors demonstrated this phenomenon by plotting the average Frobenius traces of thousands of elliptic curves from John Cremona's database. They found that curves with the same rank tended to produce similar wave patterns, while curves with different ranks produced visibly shifted oscillations. This was one of the first papers to show that elliptic curves, despite being defined by complex algebraic equations, display clear visual and statistical structures when viewed at scale.

Later, other researchers built upon this idea using machine learning to test whether these patterns could be detected computationally. One experiment applied logistic regression to classify elliptic curves by rank based on their Frobenius trace data. The model achieved an accuracy of about 97 percent, but this result was obtained mainly from curves with large conductors above 13,000, where the murmuration signals are stronger and more distinct.

Our project takes a different approach by focusing on the lower-conductor range between 0 and 10,000, where the patterns are weaker and less explored. We also introduce two new elements to the analysis. First, we explicitly graph and compare elliptic curves with conductors in the range of 0 to 5,000 and 5,000 to 10,000. Second, we systematically train logistic regression models on datasets that contain curves with different average conductor values using a sliding window approach.

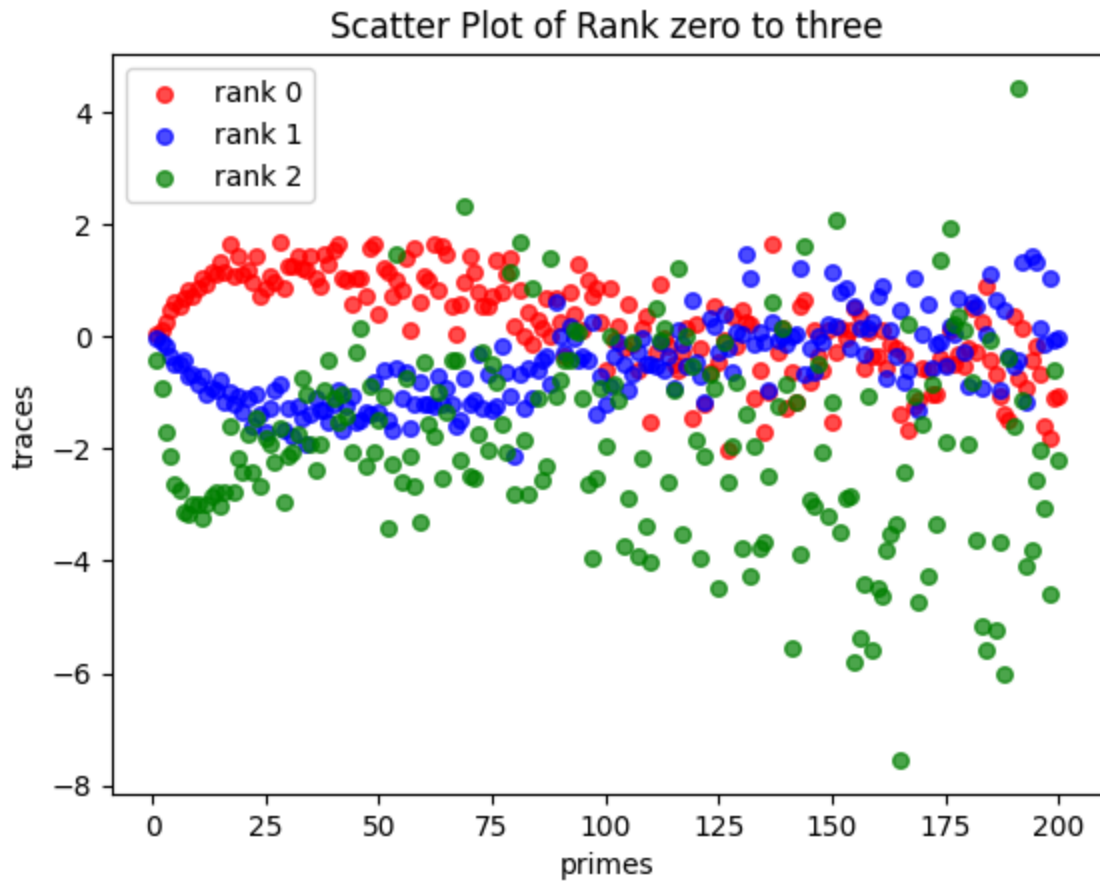
This approach allows us to study how the ability of the model to classify ranks changes as the conductor range increases. By comparing both the visual murmurations and the model accuracies, our project helps to extend the original findings of Booker and Sutherland into a new computational direction that connects number theory with data science.

4. Dataset

We originally created one dataset containing 2371 elliptic curves of conductor size 0~10,000. Every row in the dataset represents one elliptic curve. The columns in the dataset correspond to the rank and the Frobenius traces a_p for the first 200 primes p , for each elliptic curve. This dataset included 1,372 elliptic curves of rank 0, 970 elliptic curves of rank 1 and 29 elliptic curves of rank 2. This is our combined dataset. This dataset was pulled from the LMFDB, and the author computed all of the Frobenius traces

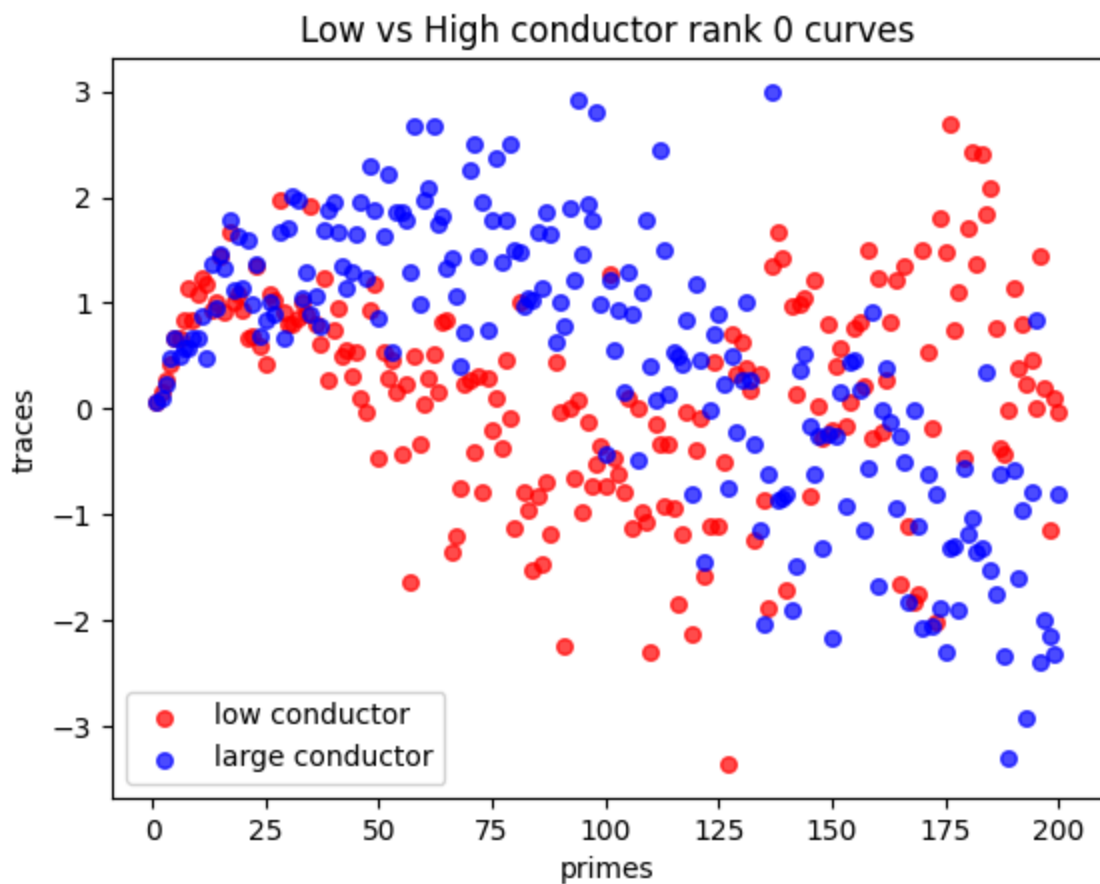
We also wanted to test logistic regression accuracy on datasets with a wide variety of conductor sizes. We first ordered by conductor (smallest to largest) all elliptic curves of rank 0. We labeled these curves from 0 to 1,372 - with the rank 0 curve of index 0 having the smallest conductor. Then for every i in the set $\{0,200,400,600,800,1000\}$, we created a dataset D_i^0 consisting of rank 0 curves with labels in the set $\{i,i+1,i+2,\dots,i+1000\}$. Then we did the same thing to create datasets D_i^1 of rank 1 curves. Finally, we created datasets D_i by concatenating D_i^1 and D_i^0 . For all of the models (including the combined model) we reserved 20% data for testing and 80% for training

For the combined dataset, D_0 , and D_{1000} , we graphed murmurations. We will explain the process for D_0 first. We began with a blank graph with the x-axis labelled “primes” and the y-axis labelled “Frobenius traces.” For each prime p , we averaged the a_p column of D_0^0 and plotted the point (p,k) in red colour where k was this average. Then we averaged the a_p column of D_0^1 and plotted the point (p,k) in blue colour where k was this second average. We did the same thing for the other datasets.



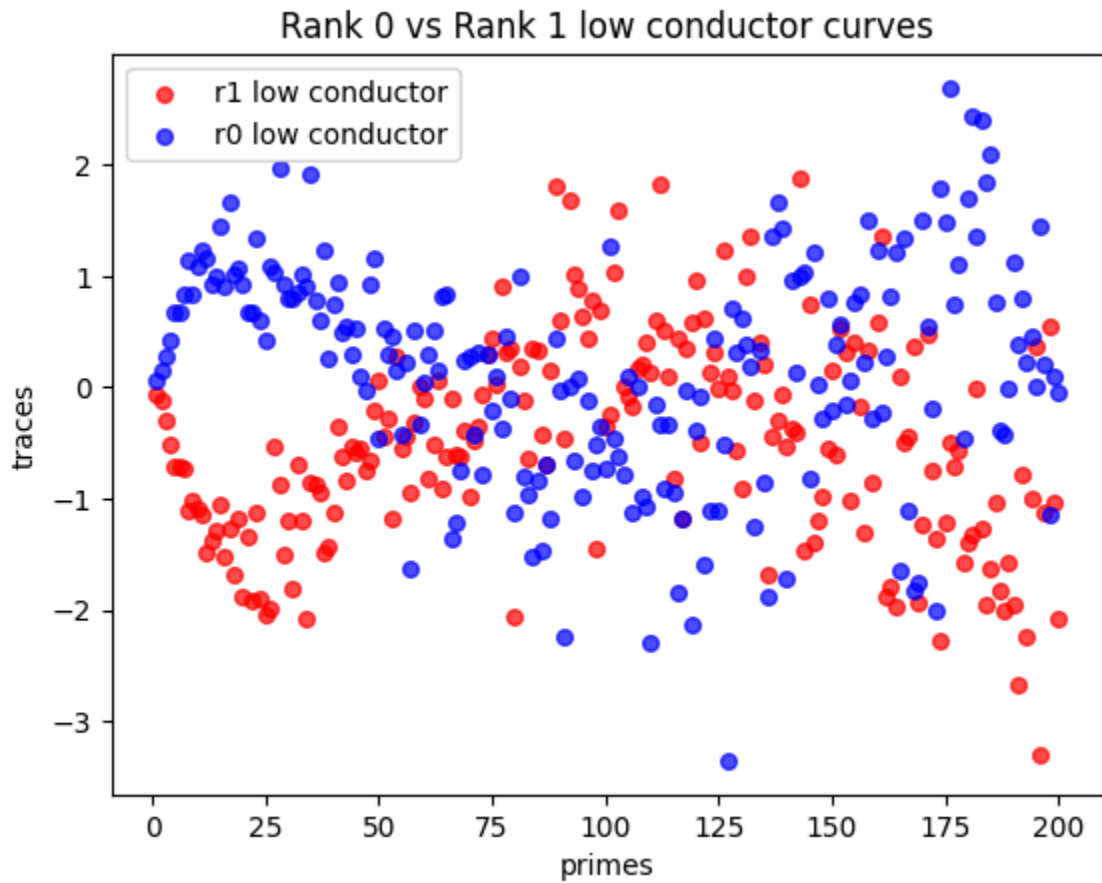
Average Frobenius traces for rank 0–2 curves

Figure 1 shows the average Frobenius traces (a_p) for elliptic curves of rank 0–2 across the first 200 primes, illustrating that the data contains measurable differences among rank.



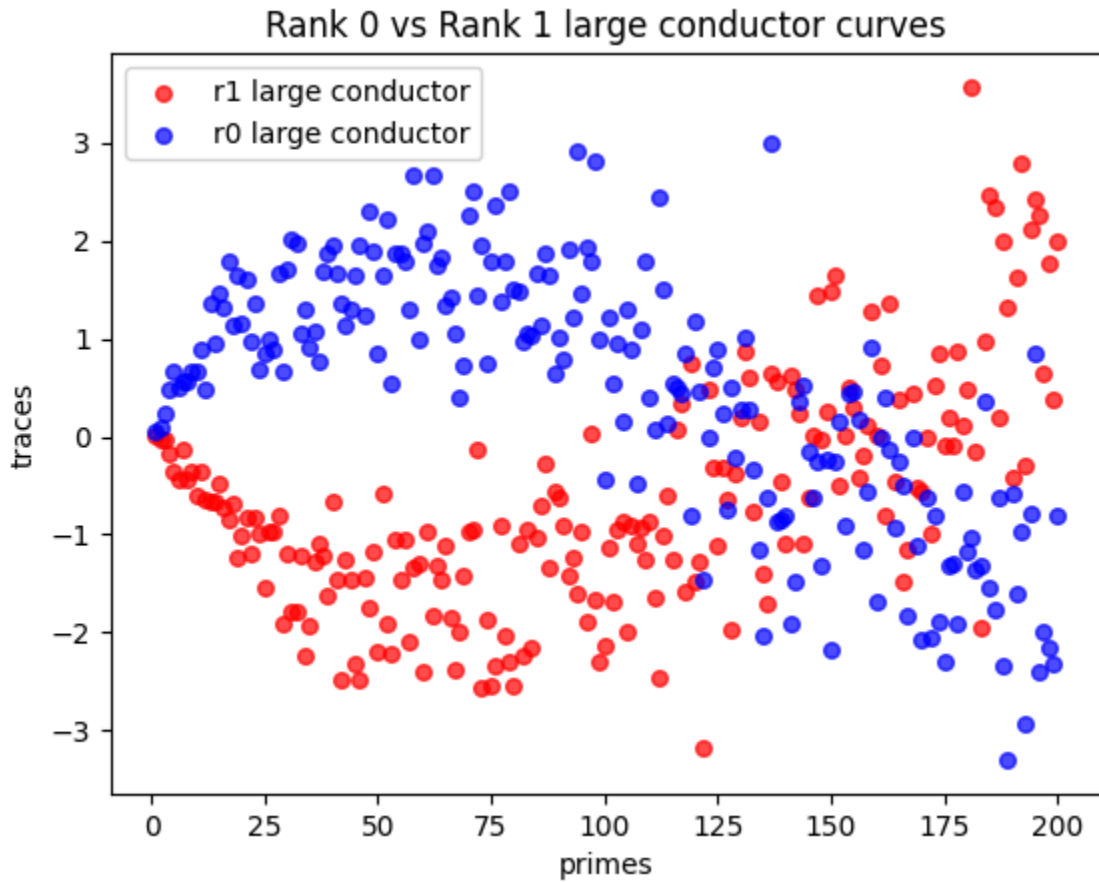
Variation in traces with conductor size

Figure 2 compares low- and high-conductor rank 0 curves, showing how trace variability increases with conductor size. Rank 1 curves show the same trend, with higher-conductor samples exhibiting broader fluctuations in a_p .



Comparison of rank 0 and rank 1 curves with low conductors.

Figure 3 compares average Frobenius traces a_p for low-conductor rank 0 and rank 1 curves across the first 200 primes. The two ranks follow similar general patterns but differ in magnitude, with rank 1 curves showing slightly greater fluctuations in a_p values.



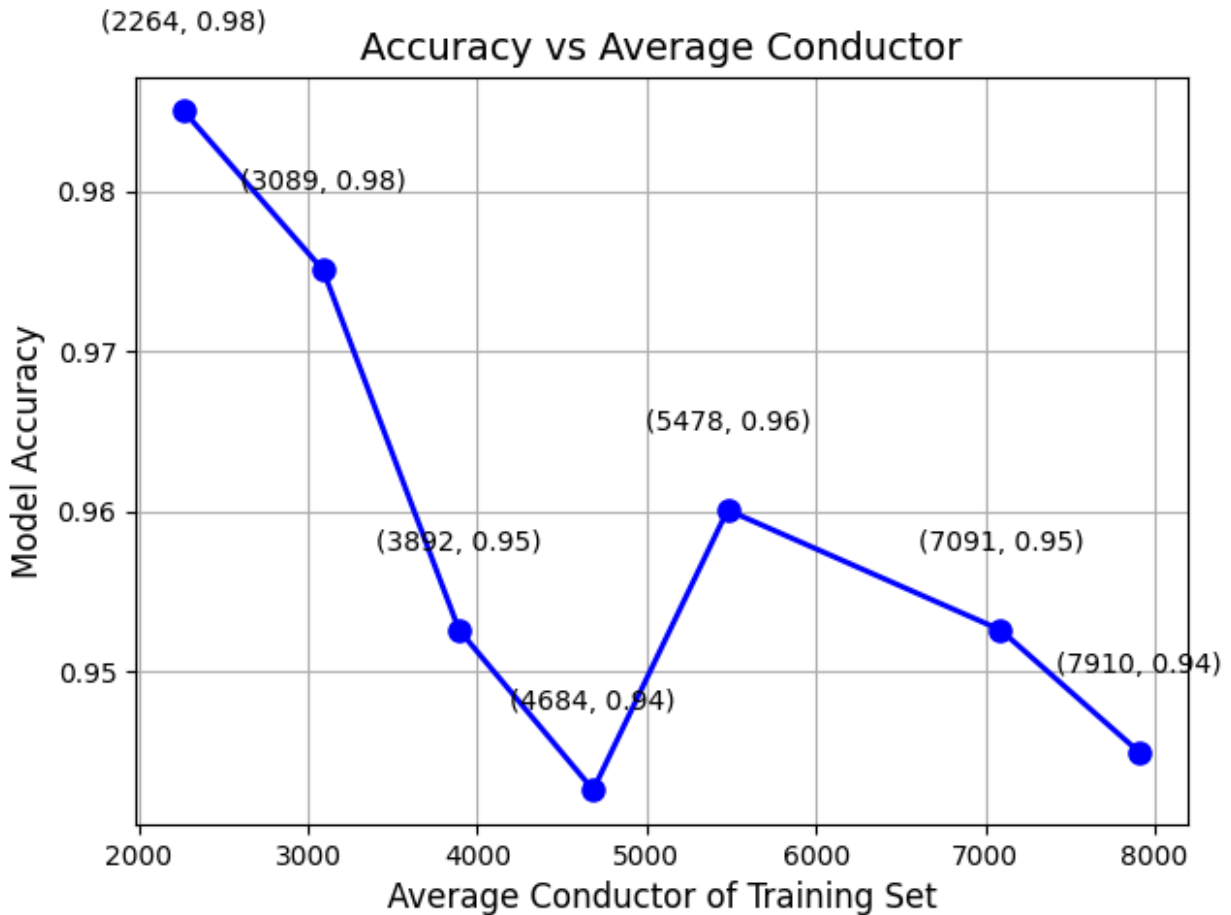
Comparison of rank 0 and rank 1 curves with high conductors.

Figure 4 compares average Frobenius traces a_p for high-conductor rank 0 and rank 1 curves. Both ranks display increased variability in traces compared to low-conductor samples, and the separation between the two ranks becomes less distinct as conductor size grows.

5. Methodology/Models

First, we built a logistic regression model trained on our entire combined dataset. Then we built 6 more logistic regression models labelled M_i for i in the set $\{0, 200, 400, 600, 800, 1000\}$. Each model M_i was trained using the dataset D_i

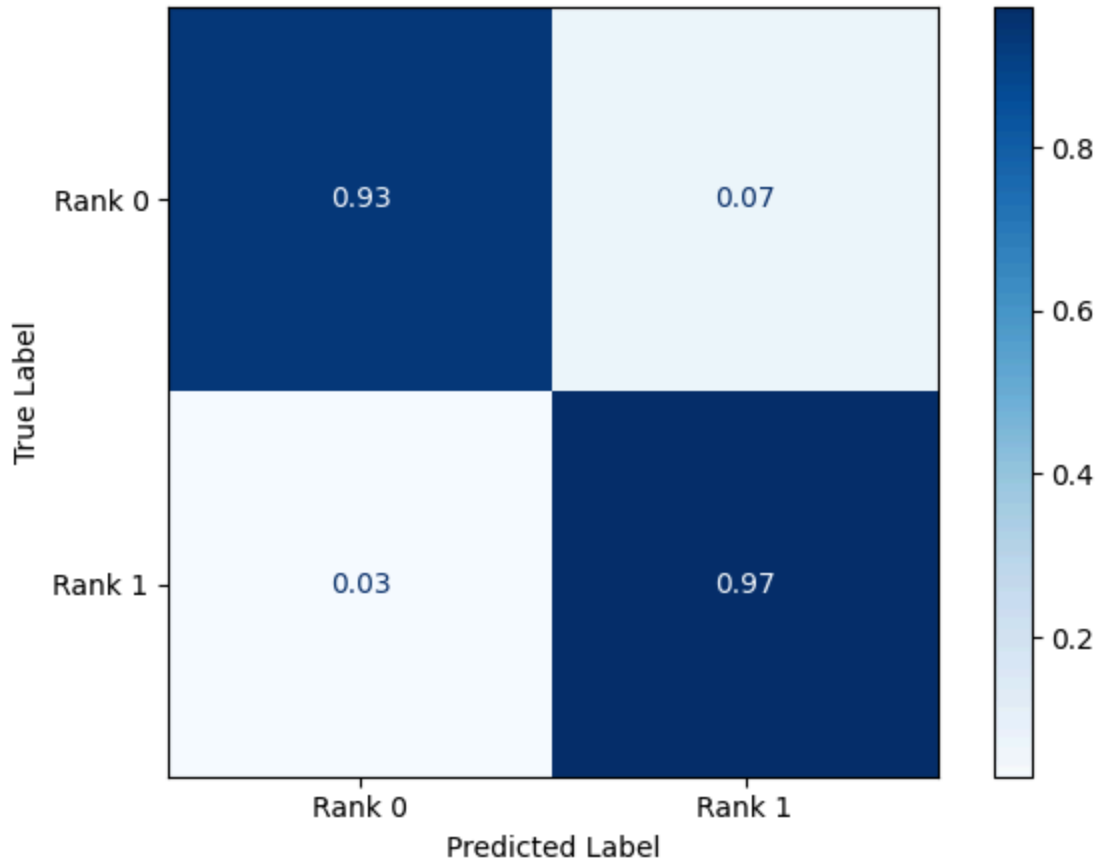
6. Results and Discussion



Model accuracy across conductor ranges

Model performance demonstrates a clear dependence on conductor size. Models trained on lower-conductor subsets achieve higher accuracy, approximately 98%, while those trained on higher-conductor subsets decrease to around 94–95%. As the average conductor increases, accuracy declines overall, reaching its lowest values in the mid-range before a modest recovery at the upper end. The relationship between accuracy and conductor is therefore negative but not strictly linear. These results indicate that curves with larger conductors possess weaker or noisier rank-separating signals for a linear classifier, likely due to increased arithmetic complexity and heterogeneity within the Frobenius traces.

Normalized Confusion Matrix for Logistic Regression (Rank 0 vs Rank 1)



Normalized confusion matrix for logistic regression (rank 0 vs rank 1).

Most predictions lie along the diagonal, indicating strong overall performance for both ranks. The model correctly classifies the majority of samples in each category, though a small portion of rank 1 curves are misclassified as rank 0. This asymmetry suggests that higher-conductor curves exhibit weaker or noisier trace patterns that reduce the model's ability to distinguish between ranks.

7. Conclusion

This study examined whether arithmetic data alone can reveal rank structure in elliptic curves. Using Frobenius traces for the first 200 primes from the Cremona and LMFDB dataset, we built a numerical corpus and trained logistic regression to classify rank 0 and rank 1 curves. The model performed strongly overall, with accuracy in the mid to upper nineties, but performance declined as conductor size increased, consistent with weaker and noisier rank signals at higher conductors. The normalized confusion matrix showed balanced precision and recall with slightly more errors for rank 1, which aligns with greater variability in its trace patterns. These results indicate that Frobenius traces carry a genuine signal for rank, yet arithmetic complexity limits linear separability.

Two constraints shaped our analysis. First, we restricted experiments to ranks 0 and 1 because rank 2 curves were too scarce for reliable evaluation. Second, our feature window covered only the first 200 primes. Future work will expand the dataset with more rank 2 examples to test whether these patterns persist beyond the binary setting, and extend the feature horizon to additional primes to study longer murmurations and their effect on model stability and accuracy. Exploring nonlinear models, stronger validation, and added arithmetic features will help clarify how conductor growth and feature complexity affect rank separability.

8. Acknowledgements

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9. References

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